SECURITY ENGINEERING RESEARCH POSTER

Version-Aware Normalization of Findings Against MITRE ATT&CK v19

Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

github.com/poojakira/attack-v19-core - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

165	63.83%	113	11
Tests passed	Statement coverage	Test functions	Test files

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Download/validate ATT&CK STIX data with SHA-256 verification.** →

02 **Parse techniques/tactics/platform metadata into typed structures.** ↓

04 **Build in-memory indexes for ID, tactic, platform, and keyword lookup.** ↓

03 **Resolve deprecated/revoked IDs through the v19 remap table.** →

05 **Regression-test migration and data-loader behavior across supported Python versions.**

Research Contribution

A typed ATT&CK v19 data layer that normalizes findings across version churn by resolving revoked/renamed technique IDs through a maintained remap table over SHA-256-verified STIX data, with indexed lookup for downstream security tooling.

Verified Evidence

EV-01	Revoked technique IDs: 22	EV-02	Total remaps: 29
EV-03	New techniques in documented v19 migration: 48	EV-04	165 tests passed.
EV-05	63.83% statement coverage; CI gate is 60%.	EV-06	113 test functions across 11 test files; parametrization expands these to 165 executed tests.
EV-07	Mapping evidence at this snapshot retains:	EV-08	22 revoked technique IDs
EV-09	29 total remaps	EV-10	48 new techniques for the documented v19 migration surface

Method and Control Path

01 **Download/validate ATT&CK STIX data with SHA-256 verification.**

02 **Parse techniques/tactics/platform metadata into typed structures.**

03 **Resolve deprecated/revoked IDs through the v19 remap table.**

04 **Build in-memory indexes for ID, tactic, platform, and keyword lookup.**

05 **Regression-test migration and data-loader behavior across supported Python versions.**

Limitations and Residual Risk

- ATT&CK mapping/normalization does not prove an attack occurred.
- O(1) dictionary/index lookup describes the in-memory data structure, not end-to-end system latency.
- Future ATT&CK releases require new migration/remap evidence.

Reproducibility

```
Expected evidence at the cited snapshot: 165 passed,
63.83% coverage. Poster evidence snapshot:
22cc9ee3e08406a446bf500f84ea1d8fb53a6a6d Poster evidence
CI run: 36783662812
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity